

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

6th September 2016

Subject CT3 – Probability & Mathematical Statistics

Time allowed: Three Hours (10.30 – 13.30 Hrs.)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception*
- 2. Mark allocations are shown in brackets.*
- 3. Attempt all questions, beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
- 4. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** Assume that a company has 99 employees and they are drawing the same salary.
- i)** Calculate the arithmetic mean and the standard deviation of salaries of these employees. (1)
- Later, one more employee has joined the company and his salary is Rs 1000 more than those of the existing employees.
- ii)** Calculate the arithmetic mean and the standard deviation of salaries of the 100 employees and comment. (3)
[4]
- Q. 2)** It is known that the value of two observations 3 and 5 have frequencies f_1 and f_2 respectively. Show that the variance of these observations is $\frac{4f_1f_2}{(f_1+f_2)^2}$. [3]
- Q. 3)** In an arrangement, a group of nine persons sit in a row at random and A, B and C belong to this group.
- Calculate the probability that
- i)** A, B and C sit together in a particular order. (2)
 - ii)** A, B and C sit together in any order. (2)
 - iii)** A and C occupy the end seats. (2)
 - iv)** B always occupies the middle seat in the row. (2)
[8]
- Q. 4)** The results of an experts committee on an investigation of car accidents, only due to tyre burst or collision with road divider, are summarized below.
- $P(\text{Accidents due to tyre burst}) = 0.6$
- $P(\text{Accidents due to collision with road divider}) = 0.4$
- It is known from past records that the death casualty in car accidents due to tyre burst and on collision with road divider are 30% and 50% respectively.
- i)** Calculate $P[\text{Tyre burst/a death casualty due to car accident}]$ (4)
 - ii)** Comment on the most probable cause of death, between the two, in car accidents. (2)
[6]

- Q. 5)** A secretary is given 100 computer passwords and only one which is correct opens a file. Since the secretary has no information on the correct password, she tries to open using one of the passwords. She randomly chooses one and discards it if incorrect until she finds the correct one.

i) Calculate the probability that she obtains the correct password in the third attempt. (2)

A security system has been set up so that if three incorrect passwords are tried before the correct one, the computer file is locked and access to it is denied.

ii) Calculate the probability that the secretary will gain access to the file. (2)

The secretary selects a password tries it and if it does not work, puts it back with the other passwords before randomly selecting a new password.

iii) Calculate the probability that the correct password is found on the tenth attempt. (2)
[6]

- Q. 6)** The random variable X has probability density

$$f(x) = \begin{cases} 8x & \text{if } 0 < x < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

and the random variable Y is such that the conditional density of $Y/X = x$

$$\text{is } f_{Y/X}(y/X = x) = \begin{cases} \frac{1}{x} & \text{if } 0 < y < x \\ 0 & \text{otherwise} \end{cases}$$

Find

i) The joint distribution of (X, Y) (1)

ii) The marginal distribution of Y (2)

iii) The mean and variance of Y , using (ii) above (4)
[7]

- Q. 7)** A random sample $X_1, X_2, \dots, X_{16}$ is drawn from a normal distribution with mean 1 and variance σ^2 . Let $\bar{X}$ and S^2 denote the sample mean and variance respectively.

Find the approximate value of $P[(\bar{X} - 1) > S]$ by referring to statistical tables. [3]

- Q. 8)** The cost X to produce a 30 second TV advertisement is a random variable with mean INR 20,000 and standard deviation of INR 3000. Suppose that $X_1, X_2, \dots, X_{100}$ is a random sample of costs to produce 100 such TV advertisements.

i) Find $E(\bar{X})$. (1)

ii) Find the standard error of $\bar{X}$. (1)

iii) Compute $P(\bar{X} < 20,300)$. (2)
[4]

- Q. 9)** It is desired to test $H_0: \theta = 1$ against $H_1: \theta = \frac{1}{2}$ based on a random sample of size 1 from an exponential distribution with density given by

$$f(x) = \theta e^{-\theta x}; x, \theta > 0.$$

The critical region is given as $x > -\log \alpha; 0 < \alpha < 1$

Show that **i)** $P(\text{Type I error}) = \alpha$ (2)

ii) Power of the test $= \sqrt{\alpha}$ (2)
[5]

- Q. 10)** **i)** Define the following terms:

- Estimator
 - Unbiasedness
 - Mean Square Error
 - Consistency
- (4)

ii) Mention any two methods of estimation. (1)

iii) Trace the steps for finding the MLE in straightforward two parameter cases. (2)

iv) Explain briefly the importance of the CRLB, in term of efficiency and confidence interval (2)

v) The claims arising from a certain portfolio of a general insurance company are believed to have a Pareto distribution with parameters α and $\lambda = 60,000$. The value of α is unknown and is to be estimated using data from a random sample of 10 claims. The amount, in INR thousands, paid by the insurer on eight claims are given below:

25.4	17.3	21.1	26.1	15.1	16.9	24.5	14.8
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The two remaining claims were known to be greater than INR 30,000 (and were passed onto a reinsurer).

Find the maximum likelihood estimate of α . (7)
[16]

- Q. 11)** A Seismologist collected data on one of the islands of Japan. The island was struck with one earthquake in past one year. She is interested in μ , the average number of earthquakes per annum.

i) Obtain an exact 95% confidence interval for μ using one year data. (4)

She also collected past data and found that the island was struck with 36 earthquakes in the past 36 years

- ii) Find an approximate 95% confidence interval for μ . (2)
- iii) Compare and comment on the confidence intervals obtained in parts (i) and (ii) above. (2)
- [8]

Q. 12) In an Organization, the salaries X (in millions of INR) paid to 200 employees over the last year have been analyzed. The results are as follows:

Amount	$X < 2$	$2 \leq X < 5$	$5 \leq X < 10$	$X \geq 10$
Number of Employees	158	24	12	6

The CEO, the highest paid employee, was paid INR100 million last year. Test at the 1% level, whether an exponential distribution with mean $\frac{1}{\lambda}$ is an appropriate distribution for the salaries. You should estimate the value of λ using the method of moments. [6]

Q. 13) A chain of LUMA ice cream parlours are floated in college campuses across the metropolitan cities. A sample of 10 ice cream parlours across different colleges is selected. The daily revenue Y (in INR) and the student population X in each of the colleges are recorded as follows.

X	200	600	800	800	1200	1600	2000	2000	2200	2600
Y	6000	10000	9000	12000	11500	14000	16000	17000	15000	20000

$$\bar{X} = 1,400; \bar{Y} = 13,050; S_{XX} = 5,680,000; S_{YY} = 156,225,000 \text{ and } S_{XY} = 28,500,000$$

The proprietor of the parlour chain has contacted a statistician, provided the above information, and asked her to predict daily revenue given the student population in a college. The statistician has decided to fit a linear regression model with student population as independent variable.

- i) Fit a linear regression model. (2)
- ii) Find the 99% confidence interval for the slope parameter. (3)
- iii) Establish statistically, based on the regression model (i) above, whether there is any relationship between the daily revenue and the college student size by proposing and testing an appropriate hypothesis (2)
- iv) Find the 95% confidence interval for the mean daily revenue when the population of the college is 1,000. (3)
- v) For what size of the college student population, the 95% confidence interval for the mean daily revenue is the shortest? (2)
- vi) Find 95% confidence interval for the predicted daily revenue for an individual parlour when the student population of one particular college is 1,000. (2)
- vii) Interpret the results of (iv) and (vi) (1)
- [15]

- Q. 14)** Three different insurance companies, (1 – General; 2 – Health; 3 – Life) were asked to provide data regarding the salaries (in lakhs of INR p.a.) of entry level actuarial analysts from the respective HR departments. The results were as follows:

Company1	6	5	4	3	4		
Company2	4	6	5	3			
Company3	5	6	5	4	3	4	3

Consider the model:

$$Y_{ij} = \mu + \tau_i + e_{ij}; \quad i = 1, 2, 3; j = 1, 2, 3 \dots n_i; e_{ij} \sim N(0, \sigma^2)$$

Where Y_{ij} is the j th result of the i th company, n_i is the number of responses available for company i , the e_{ij} are independent and $\sum_i n_i \tau_i = 0$.

- i) Calculate the least squares estimates of μ and τ_i ; $i = 1, 2, 3$. (4)
 - ii) Analyze the above data using least significant difference approach. On the basis of your analysis, among these three, which company would you suggest a fresher to apply for a job? (5)
- [9]**
